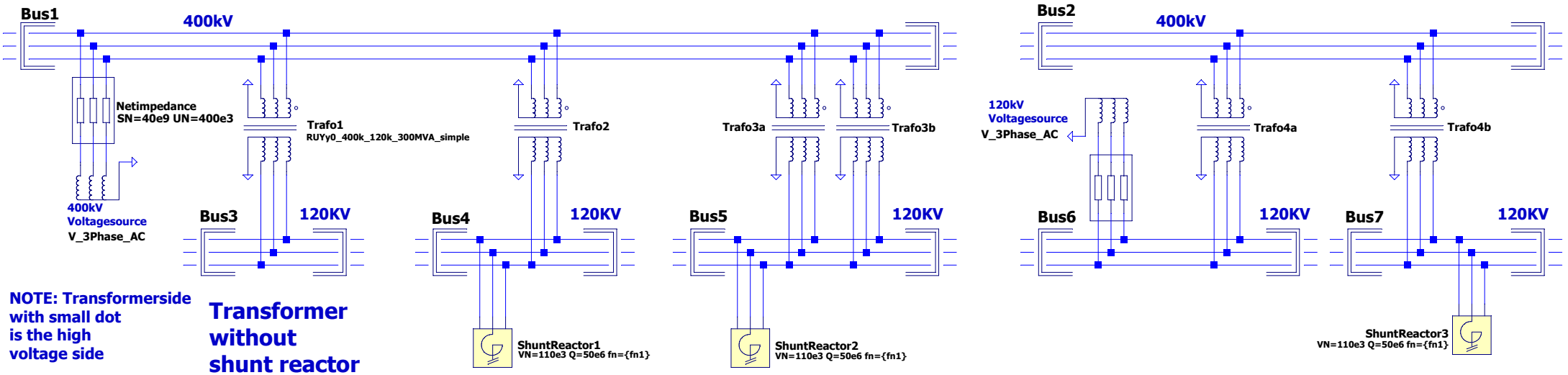
# Transformer and shuntreactor AC-Analysis

.ac list 49.999 50 50.001

```
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1} ; should always be present
.options numdgt=10 plotwinsize=0 ; for accuracy .meas statements
.options vntol=1m abstol=1u ; for accuracy voltage and current
.options meascplxfmt=polar ; for meas AC values in polar format
*.options logparams ; To put input parameters of all devices to the log file (Version>24.x)
```




NOTE: Transformerside with small dot is the high voltage side

**Transformer without shunt reactor**

**Transformer with shunt reactor**

**Two transformers in parallel and shunt reactor**

**Two transformers in series and shunt reactor**

See the measurements:
Activate schematic window
View->Spice Output log
Scroll to the end of the file

```
; Reactive powers: (at Shuntreactor 1,2,3)
.meas AC QSR_1  FIND 3*Ix(ShuntReactor1:L1)*v(bus4a:L1) at 50
.meas AC QSR_2  FIND 3*Ix(ShuntReactor2:L1)*v(bus5a:L1) at 50
.meas AC QSR_3  FIND 3*Ix(ShuntReactor3:L1)*v(bus7a:L1) at 50
; Voltages: (at Shuntreactor 1,2,3)
.meas AC VSR_1  FIND v(bus4:L1)  at 50
.meas AC VSR_2  FIND v(bus5:L1) at 50
.meas AC VSR_3  FIND v(bus7:L1) at 50
; Currents: (at Shuntreactor 1,2,3)
.meas AC ISR_1  FIND Ix(ShuntReactor1:L1) at 50
.meas AC ISR_2  FIND Ix(ShuntReactor2:L1) at 50
.meas AC ISR_3  FIND Ix(ShuntReactor3:L1) at 50
```

To see the simulation voltage results simply click on the lines
To see the currents click on the input pins of the device
The left buses measurement pins shows the line to line voltages
The right buses measurement pins shows the line to ground voltages